



FA report of TP NG issue

JUNGLIM G140HAB01.2

Problem Statement

- General Information

Brand Customer	JUNGLIM	Defect Phenomenon	TP NG
SI Customer		Defect Rate or Q' ty	1 pcs
AUO Model	G140HAB01.2	Defect Station	Line Reject
AUO S/N	D345B2K4GSZZS0100	RMA No.	762612100246
Array/Cell/Beol/Module	S01	Chip ID	
Week Code	2508	Product Provider	YH Chuang
FMA analysis grade		Issue Date	2026/02/10

- Defect Panel S/N Picture



- History

-

Problem Statement

- Describe the phenomenon

- JUNG LIM identified one unit of G140HAB01.2 panel with TP NG failure.
- The reported symptom was verified and can be reproduced under AUO inspection conditions.

- Defect information

Windows Studio ITS 1.0.11.4 RELEASE

Module_Mode Profile Name : Profile\1112532000140900_V7009_0002_FD_20240821.ini

Length: 0

Profile Load Report

FAIL

PASS	0
FAIL	1
TOTAL	1
YIELD	0%

Test Name	SPEC	Status	Progress	View Report
FW_Verify	FW CheckOnly Path = 1112532000140900_V7009_0002_AUD_0x54CF_0x95DA.hex	Pass	100%	FW Version : 7.0.0.9.0.0.0.2 CRC : F506,C550,40E8,AD50,F526,F725,
IC_Verify	IC_Type = 1112532 Interface = HID PID = 0x54CF KChannel = 74 YChannel = 42	Pass	100%	Click for result
Short_Test	Max_Threshold = 300	Fail	100%	Click for result
Open_Test	Min_Threshold = 630	Fail	100%	Click for result
Uniformity_Test	Uniformity RawData En Uniformity 2ndiff En Uniformity 3rdiff En	Fail	100%	Click for result
Test Result (8.94sec)		Fail	0	Click for result

TP NG

Root Cause Analysis

- Step 1: Visual Inspection:

No abnormality was found on the panel appearance
No cosmetic defect, damage, or contamination was observed on either front or back side



- Step 2: Functional Verification:

The TP NG issue reported by customer was rechecked at AUOTest results confirmed failure at Y23 channel in Open & Short Test Detailed log data is provided for reference

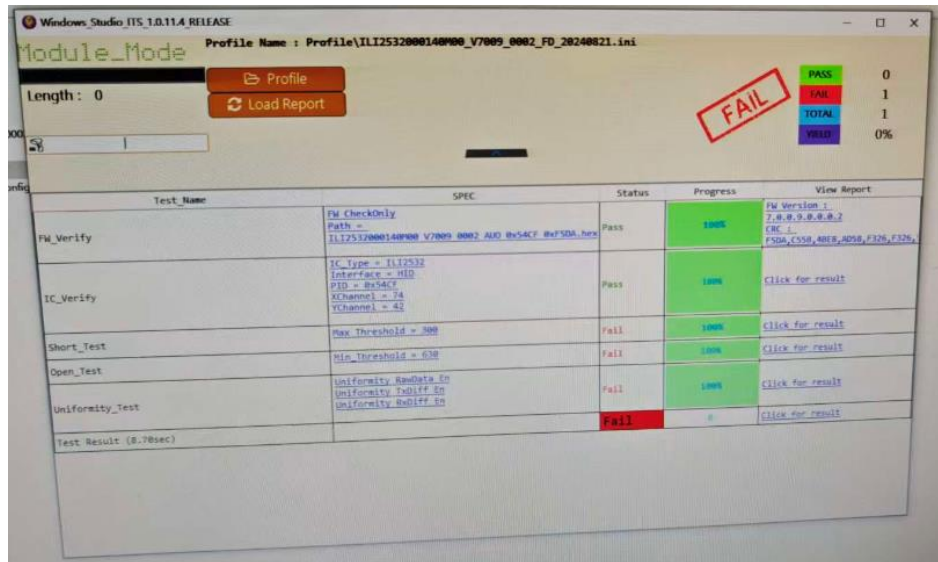


NG-SensorTestL
OG-12313231212:

Root Cause Analysis

Step 3: Lens Removal Verification

- After removing the lens, TP was re-tested
- The failure remains unchanged and is still observed at Y23 channel in Open & Short Test.
- Test results are consistent with the previous findings.



Test_Name	SPEC	Status	Progress	View Report
FW_Verify	FW CheckOnly Path = IL12532000140000_V7009_0002_AUO_8x54CZ_8x730A.hex	Pass	100%	FW Version 1 2.0-0.0-0.0-0.2 CRC 1 F50A,C558,40E8,4050,F326,F320,
IC_Verify	IC_Type = IL12532 Interface = HID PID = 8x54CZ XChannel = 34 YChannel = 42	Pass	100%	Click for result
Short_Test	max_Threshold = 300	Fail	100%	Click for result
Open_Test	Min_Threshold = 630	Fail	100%	Click for result
Uniformity_Test	Uniformity_RmdData_E0 Uniformity_RmdDiff_E0 Uniformity_RmdDiff_E0	Fail	100%	Click for result
Test Result (8.70sec)		Fail	0	Click for result

Results are consistent
with previous test logs

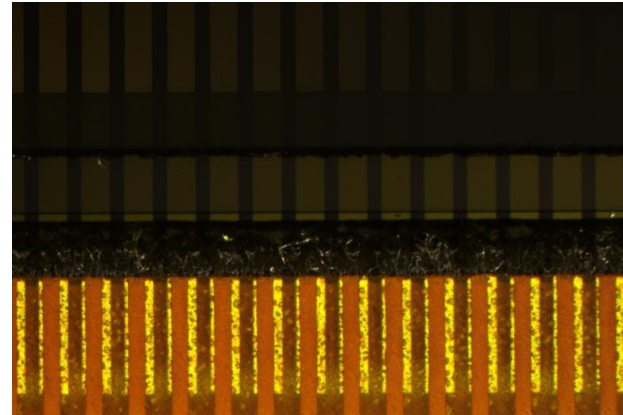
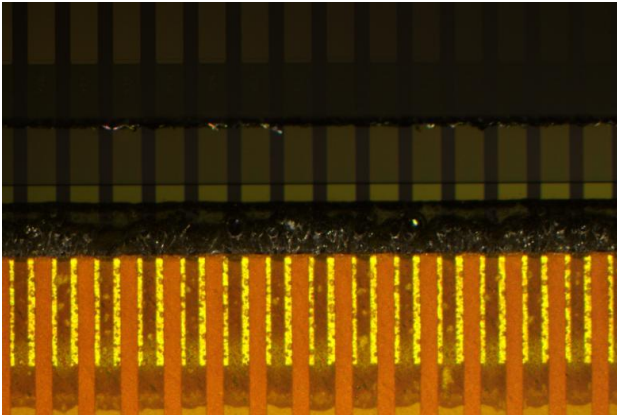
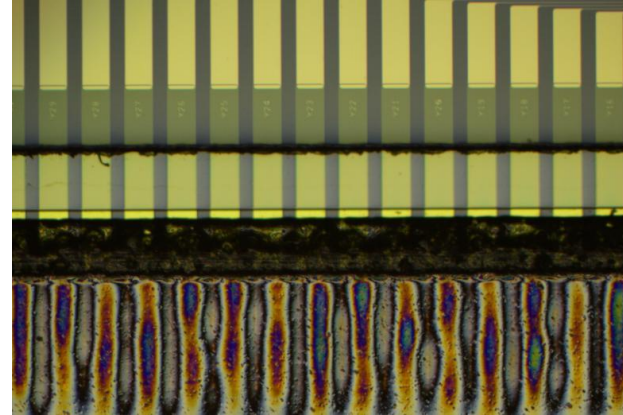
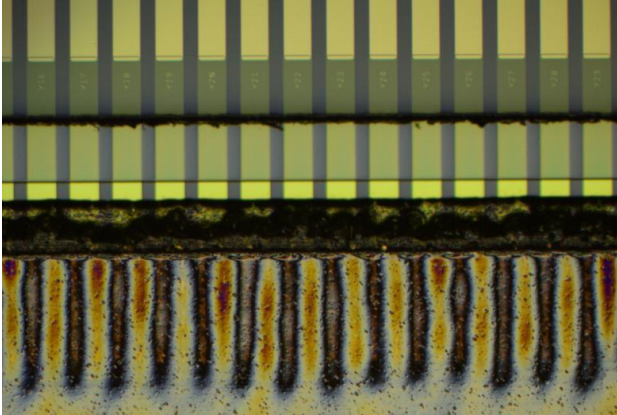


NG-SensorTestL
OG-23112312321:

Root Cause Analysis

Step 4: OM Inspection of TP FPC Bonding

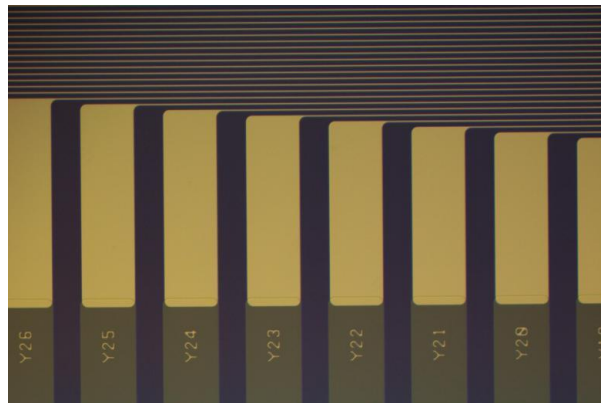
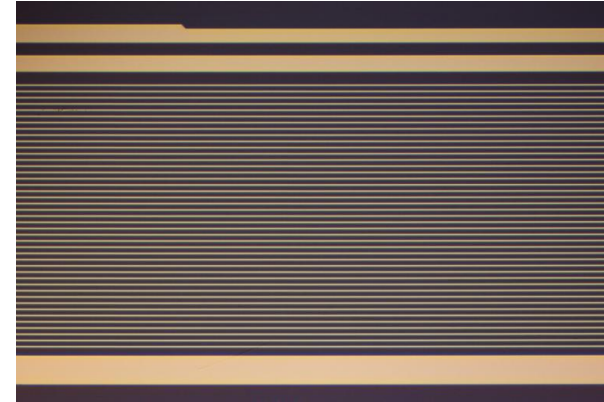
- Optical microscopy (OM) inspection was performed on TP FPC bonding area
- No abnormality was observed in the bonding interface
- No open, short, delamination, or misalignment was identified;



Root Cause Analysis

Step 5: TP Circuit Inspection (OM)

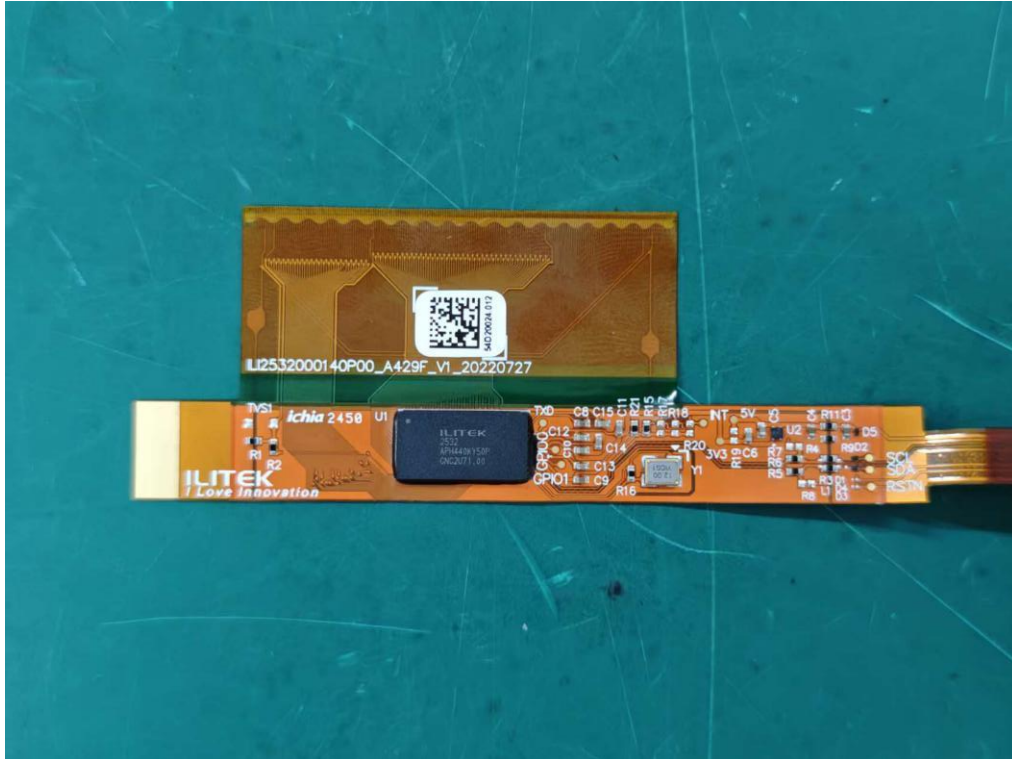
- Optical microscopy inspection was performed on TP routing lines
- No abnormality was observed in the circuit pattern
- No open, short, breakage, or deformation was identified;



Root Cause Analysis

Step 6: TP FPCa Appearance Inspection

- Visual inspection was performed on TP FPCa
- No abnormality was observed on the FPC or component area
- No damage, contamination, or missing components were identified



Panel analysis (AUO) Summary

- 1.The TP NG issue reported by customer was confirmed by AUO re-test, showing failure at Y23 channel in Open & Short Test
 - 2.Based on the analysis results, no abnormality was found in panel, bonding, or TP circuit routing
 - 3. The issue is initially considered to be related to TP FPCa
- ADP(AUO) request our supplier Ichia to do further analysis



Part No.	Mfg Plant	Part Description	Material Group Name	Where Used	Status	VQA Maker BU	Maker Group	Maker↓
54D20024.012		FPC ASSEMBLY G140HAB01 G140HAB01.2 / ILI2532 / USB / 5V	SMT (FPCA)	View	RELEASE	GD	D0206- 毅嘉	M00272- 毅嘉電子(蘇州)有限公司

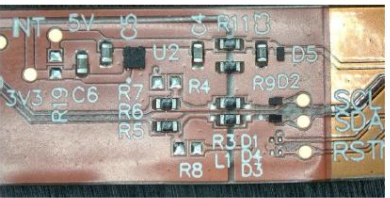
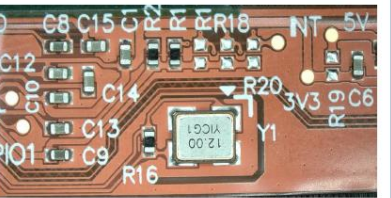
FPCa analysis (Ichia)

- According to supplier (Ichia) analysis, no abnormality was found in FPCa appearance and soldering condition
- All inspected areas, including IC, components, and solder joints, are confirmed to be normal
- No visible defect related to the failure was identified

ichia

真因分析与验证 Root Cause Analysis & Verification

步驟1：外观确认：检查客诉品焊点及外观未见不良。

狀態	外观检查		
图片			
			
说明	检查客诉品焊点及外观未见不良。		

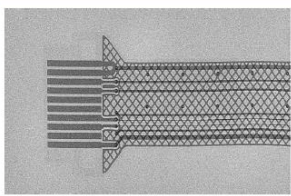
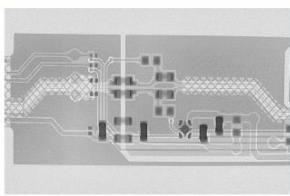
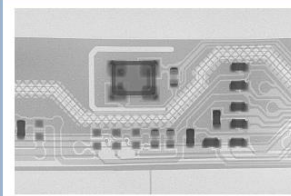
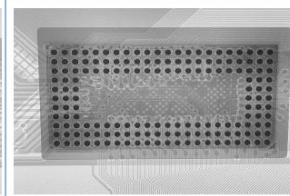
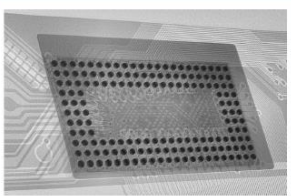
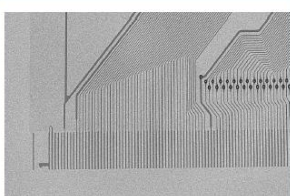
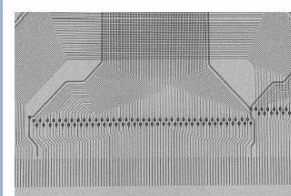
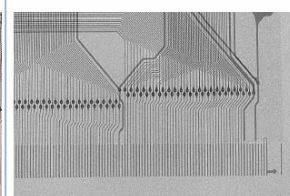
FPCa analysis (Ichia)

- According to supplier (Ichia) X-ray analysis, no abnormality was found in solder joints or internal circuit structure
- All inspected areas show normal connection condition without open, short, or void-related defects
- No physical defect was identified that could explain the failure

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真因分析与验证 Root Cause Analysis & Verification

➤ 步骤2：X-ray确认：X-Ray扫描客诉品焊点及线路无异常。

状态	X-Ray检测			
	图片			
说明				
				
说明		X-Ray扫描客诉品焊点及线路无异常。		

FPCa analysis (Ichia)

- Ichia re-tested the returned sample and successfully reproduced the failure
- The TP NG condition is consistent with customer feedback, showing abnormality at Y23 channel
- Test results align with AUO findings

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真因分析与验证 Root Cause Analysis & Verification

➤ 步驟3：对客诉制品进行复测：客诉品复测Fail,显示Y23不良与客户反馈一致。

状态		复测确认				
图片						16 (Test Result) 17 Normal (X_1) (X_2) (X_3) (X_22) (X_23) 18 X_SLK 5 3 5 4 2 19 (Y_1) (Y_2) (Y_3) (Y_22) (Y_23) 20 21 Y_SLK 3 2 2 7 *32760 22 23 24 25 ===== 26 Result_Summary 27 28 {FW_Verify} OK 29 {IC_Verify} OK 30 {Short_Test} NG 31 =====
说明	客诉品复测Fail,显示Y23不良与客户反馈一致					

FPCa analysis (Ichia) Summary

1. Production records indicate all ICT and FCT tests were passed during manufacturing
2. No abnormality was found in appearance inspection, soldering condition, or X-ray analysis
3. The failure can be consistently reproduced at Y23 channel, matching customer and AUO findings
4. Electrical measurement confirms normal continuity and resistance of the signal path

→ Ichia request the IC supplier ILITEK to do further analysis

ichia

结论 Conclusion

➤ 分析结果 Analysis Result:

1. 生产信息确认(当时ichia厂内ICT、FCT测试都为PASS)
2. 外观确认：检查客诉品焊点及外观未见不良。
3. X-ray确认：X-Ray扫描客诉品焊点及线路无异常。
4. 复测检测：客诉品复测Fail,显示Y23不良与客户反馈一致。
5. 量测确认：客诉品Y23对地阻值与OK比对无异常。

➤ 建议及可能的根本原因 Suggestion & Possible Root Cause:

综上所述，客诉品焊点及线路无异常，初步怀疑为IC不良。零件料号：FPCB381AIL1

IC analysis (ILITEK)

- Based on ILITEK analysis, no abnormality was observed on IC appearance during visual inspection
- No damage, contamination, or abnormal marking was identified on the IC surface
- The IC physical condition is confirmed to be normal

Analysis - Appearance inspection

- No obvious abnormalities were found during visual inspection.

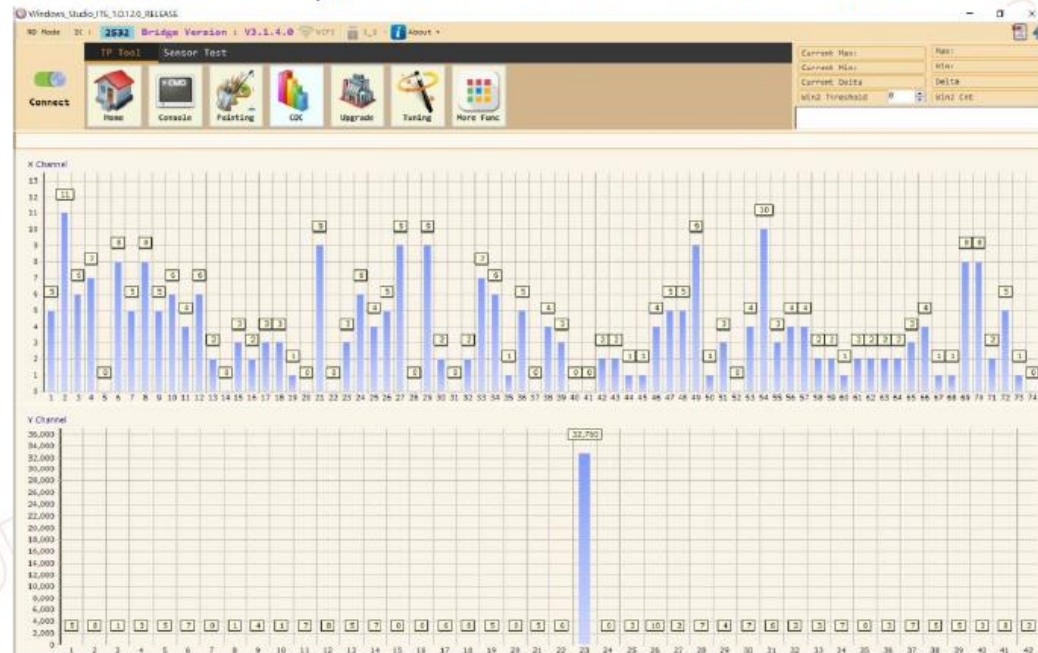


IC analysis (ILITEK)

- Based on ILITEK analysis, the TP short test result can be successfully replicated
- The abnormality is consistently observed at Y23 channel
- Test results show deviation from normal channel behavior

Analysis - Duplication phenomenon

- TP Short test results can be replicated to Y23 values, but anomalies exist.



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- | | |
|----------|--------|
| TX24-GND | 0.735V |
|----------|--------|

0.735V

Summary: It is sp

IC analysis (ILITEK) Summary

1. The NG IC behavior can be reproduced and is consistent with Y23 short test failure
2. Diode voltage measurement shows that TX23 to GND (0.659V) is lower than adjacent TX channels (~0.735V)
3. The deviation indicates abnormal electrical characteristics at TX23 channel

Summary

- NG IC can be copied to Y23 short test failed.
- The diode voltage of TX23 to ground is lower than that of the adjacent Tx pin.
We speculate that the TX23 diode is damaged, usually caused by abnormal voltage or current.

Overall

1. TP NG issue has been verified and can be consistently reproduced at Y23 channel
2. AUO analysis confirms no abnormality in panel, bonding, or TP circuit routing
3. Supplier (Ichia) investigation further verifies no defect in FPCa structure, soldering condition, or electrical continuity
4. The failure is localized, stable, and reproducible, and is not related to manufacturing or structural defects

Technical Findings

1. The abnormality is consistently associated with the Y23 / TX23 channel
2. IC analysis shows channel-specific electrical deviation at TX23
3. Diode voltage measurement indicates TX23 is lower than adjacent channels, showing abnormal electrical characteristics

Final Statement

No physical or structural defect was identified across panel, FPCa, or IC manufacturing quality. The issue is considered to be related to electrical overstress (EOS) impact on the TX23 channel, leading to abnormal channel behavior.



Tap Into The Possibilities

AUO